

Sl. No. of Ques. Paper : 5720
Unique Paper Code : 234405
Name of Paper : (CSHT-410) Software Engineering
Name of Course : B.Sc. (Hons.) Computer Science
Semester : IV
Duration : 3 hours

F

Maximum Marks : 75

(Write your Roll No. on the top immediately on receipt of this question paper.)

The paper has two Sections. All questions in Section A are compulsory. Attempt any four questions from Section B. Subparts of the questions should be answered together.

Section A

- Q1.
- i) "Software is both a product and a vehicle that delivers a product". Justify the statement 3
 - ii) What is Rapid Application Development (RAD) Process Model? Under what circumstances, it is recommended. 3
 - iii) Explain software sizing in context of software Project Estimation. 3
 - iv) List the characteristics of a good SRS (Software Requirement Specification). 3
 - v) What is the significance of Gantt Chart in Project Scheduling? 3
 - vi) Explain the use of Problem Partitioning in Software Design. 3
 - vii) What do you understand by Smoke Testing? List the activities involved in Smoke Testing Approach. 3
 - viii) What types of risks are likely to encounter as software is built? 3
 - ix) Define DRE (Defect Removal Efficiency). 3
 - x) What do you understand by Project Metrics? List any two advantages of Project Metrics. 3
 - xi) What are umbrella activities? List any two umbrella activities. 3
 - xii) Explain any two measures of software quality. 2

Section B

- Q2. a) "Software Engineering is a layered technology." What are these layers and state the significance of each layer? 1+4
- b) List the generic process framework activities. 5

Turn over

- Q3. Draw a context Data Flow Diagram (DFD) and Level 1 DFD for Payroll Software. Also, develop Data Dictionary for the same. 3+4+3
- Q4. a) What is the significance of Abstraction in Software Design? Describe the two common abstraction mechanisms for software system design? 2+2
- b) What do you understand by Structured Design Methodology (SDM)? Illustrate the steps involved in SDM with the help of an example. 1+5
- Q5. a) Explain two different approaches to test a software? 4
- b) Create a flow graph to find the cyclomatic complexity of the following code. Also show the no. of independent paths and regions: 4+2
- ```

Void sum()
 Input: Array of n integers
 Output: Sumeven and Sumodd
 Begin
 i = 1, Sumeven = 0, Sumodd = 0
 while (i <= n)
 Begin
 if (a(i) % 2 == 0)
 Sumeven = Sumeven + 1
 else
 Sumodd = Sumodd + 1
 Endif
 i = i + 1
 Enddo
 End

```
- Q6. a) Illustrate and explain the determinants for Software Quality and Organizational Performance. 4
- b) Use the COCOMO II model to estimate the effort required to build software that produces 12 screens and 10 reports, and will require approximately 80 software components. Assume average complexity (screen-2, reports-5) and average developer/environment maturity as 13. Use the application composition model with object points. 6
- Q7. Differentiate among the followings (Attempt any four): 2.5x4
- Alpha and Beta Testing
  - Coupling and Cohesion
  - Direct and Indirect Measures of Software
  - Verification and Validation
  - Analysis and Design model